Python Foundations Demo

REX_SYNTHETIC_DEMO | Fictional TA example. No real academy offering, instructor, accreditation or tender.

Course ID: DEMO-PY-101

Title: Python Foundations Demo

Description: A fictional practical introduction to Python programming using variables, conditions, loops, functions and CSV files. Learners build a small expense tracker using invented transactions.

Level: Beginner

Training format: Online

Price EGP: 2400 EGP

Duration hours: 24

Department: Software Development

Instructor name: Nour Demo Instructor

Instructor bio: Fictional instructor created solely for the TA demonstration.

Schedule: Demo cohort: 8 sessions of 3 hours, Sunday and Tuesday 18:00-21:00 Cairo time. Dates require confirmation.

Offers: Demo offer: 10 percent off for a fictional group of five; this is not a real promotion.

Target audience: Beginners who want to automate simple office tasks.

Career opportunities: Foundation for junior Python development; Office task automation

Admission requirements: Access to a laptop with Python 3.11 or later; Ability to manage local files

Prerequisites: Basic computer use; no programming experience required

Learning outcomes: Write conditions, loops and reusable functions; Read CSV files and validate numeric values; Build and test an expense tracker

FAQs: [{"question": "Is programming experience required?", "answer": "No. Basic computer use is enough for this fictional beginner course."}, {"question": "Is a job guaranteed?", "answer": "No employment or accreditation promise is made."}]

Course content

This material is an authored fixture for testing ingestion and document workflows. It is not an accredited curriculum.

Getting started

Install Python and run print statements. Store an expense amount in a variable. Distinguish strings from numbers and convert input with float. Exercise: calculate a subtotal and 14 percent tax for invented items.

Decisions and repetition

Use if to reject negative amounts, for to total a list and while to repeat a menu. Exercise: process [120, 80, 50] and verify the total is 250. Explain how an empty list should behave.

Functions and files

Write add_expense(amount, category). Use csv.DictReader and handle missing or malformed amounts with a clear error. Exercise: load three fictional rows and skip an invalid row while reporting its line number.

Expense tracker project

Create a menu to add, list and total expenses. Keep data in a local CSV file. Submit the script, a README and tests for empty input, negative amounts and nonnumeric input. Assess correctness 50 percent, validation 30 percent and clarity 20 percent.